4 **(a)** Fig. 4.1 shows a circular coil of 500 turns and radius 0.12 m.

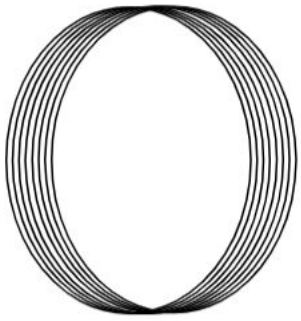


Fig. 4.1

A varying magnetic field is applied at right angles to the plane of the coil. Fig. 4.2 shows the variation with time *t* of this magnetic field of flux density *B*.

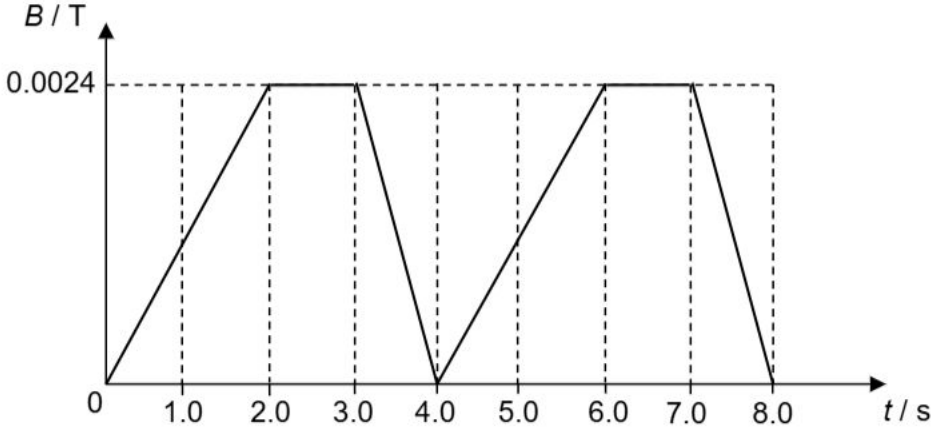


Fig. 4.2

(i) On Fig. 4.3, sketch a graph to show the variation with time t of the induced e.m.f. *E* in the coil. Include values on the y axis.

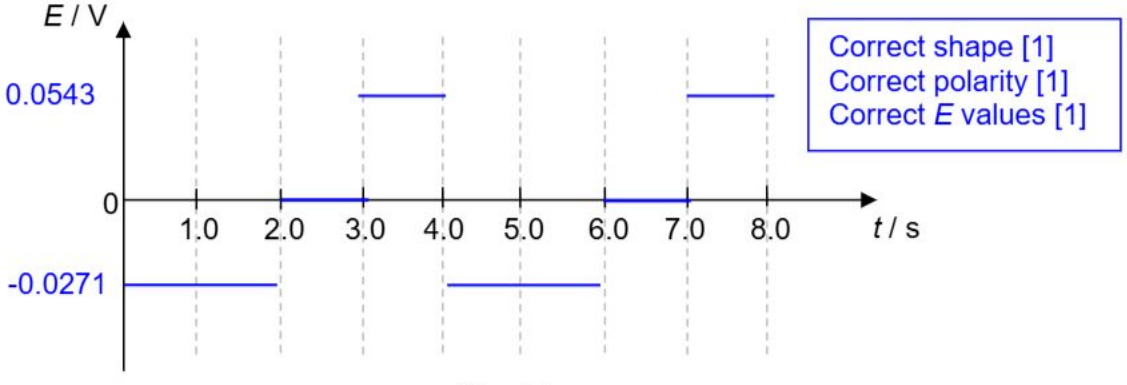


Fig. 4.3

[3]

(ii) Hence, determine the r.m.s value of the induced e.m.f.

$$\text{r.m.s.} = \sqrt{\frac{(0.0271^2 \times 2) + (0.0543^2 \times 1)}{4}} \quad \text{M1}$$

$$= 0.0332\text{A} \quad \text{A1}$$

r.m.s. value of induced e.m.f. = ... V [2]

(iii) The e.m.f. source that you have sketched in Fig. 4.3 is now connected in series with a resistor *R* and an ideal diode as shown in Fig. 4.4.

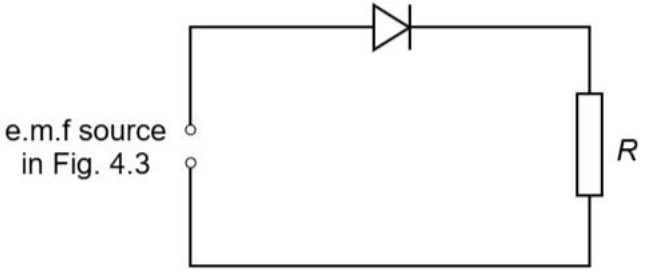


Fig. 4.4

State and explain the potential difference across R when the ideal diode has maximum potential difference across it.

When diode has maximum p.d. across it, it is reverse bias / not conducting / has infinite resistance. [M1].

The sum of the p.d. across R and diode must equal to the e.m.f. of the supply. [M1]

Since resistance of R is insignificant compared to that of diode, p.d. across R is zero [A1]. OR The total resistance of the circuit is infinite and the current in the circuit is zero. [M1] Since there is no current in R, p.d. across R is zero. [A1]

[3]

[Total: 8]